

1. First Drone Connection Script (Analysis)

This script's purpose is to open an MAV Link connection to a simulated vehicle and print three telemetry values to confirm the connection works.

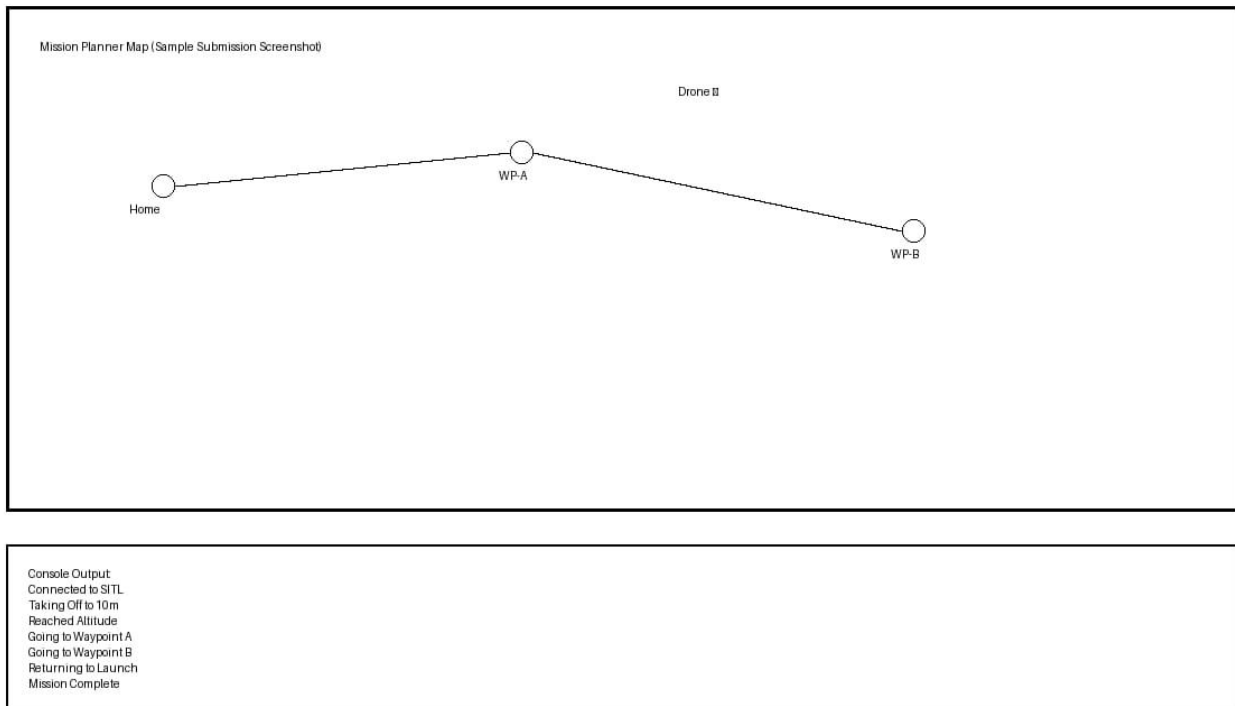
```
from drone kit import connect

# Connect to the Vehicle (SITL)
print("Connecting to vehicle...")
vehicle = connect('127.0.0.1:14550', wait ready=True)

# Print some basic information
print("Drone is connected")
print("GPS:", vehicle.gps_0)
print("Battery:", vehicle.Battery)
print ("Last Heartbeat:", vehicle. last heartbeat)

vehicle. Close ()
print("Connection closed")
```

2. Screenshot:



Line-by-line explanation

Code	Explanation
<code>from drone kit import connect</code>	Imports only the <code>connect ()</code> function from the Drone Kit library — the entry point for linking to any vehicle.
<code>connect ('127.0.0.1:14550', wait ready=True)</code>	127.0.0.1:14550 is the local loopback address and port where Ard Pilot SITL streams MAV Link data by default. <code>wait ready=True</code> pauses the script until Drone Kit has downloaded the vehicle's parameters and initial telemetry, so the values read afterward are valid (not <code>None</code>).
<code>vehicle.gps_0</code>	Returns a GPS Info object describing GPS quality (see parameter table below).
<code>vehicle. Battery</code>	Returns a Battery object with voltage, current, and remaining charge level.
<code>Vehicle. last _heartbeat</code>	Returns the time (in seconds) since the last MAV Link heartbeat was received — used to detect a lost connection.
<code>vehicle. Close ()</code>	Cleanly shuts down the MAV Link connection thread.

3. Deliverable — Explanation of Each Parameter

The task deliverable asks for a screenshot of the script's output plus an explanation of each printed parameter. The parameter-by-parameter explanation is below, followed by the expected output format and instructions to capture your own real screenshot.

3.1 GPS — `vehicle.gps_0`

This is a GPS Info object. Its sub-fields are:

- `fix type` — Type of GPS lock: 0/1 = no fix, 2 = 2D fix, 3 = 3D fix (3D is needed for reliable autonomous flight).
- `satellites visible` — Number of satellites currently being tracked. More satellites generally means better position accuracy.
- `edh` — Horizontal position error estimate (HDOP-related); lower values mean better horizontal accuracy.
- `eve` — Vertical position error estimate; lower values mean better altitude accuracy.

3.2 Battery — `vehicle. Battery`

This is a Battery object. Its sub-fields are:

- `voltage` — Pack voltage in volts. Used to judge remaining flight time and detect under-voltage.
- `current` — Current draw in amps (may show as `None` in some SITL setups since simulated current sensing is optional).
- `level` — Remaining battery percentage (0–100), or -1 if the autopilot is not reporting it.

3.3 Last Heartbeat — `vehicle. last _heartbeat`

A MAV Link heartbeat is a small "I am alive" message the autopilot sends about once per second. `last heartbeat` reports how many seconds have passed since the most recent one arrived. A small, stable

number (close to 0–1 s) confirms the link is active; a rapidly growing number means the connection has been lost.

4. Expected Output (Sample, for Reference)

```
Connecting to vehicle...  
Drone is connected  
GPS: GPS Info: fix=3, Numata=10  
Battery: Battery: voltage=12.587, current=0.0, level=100  
Last Heartbeat: 0.92  
Connection closed
```

Reading this sample: fix=3 means a full 3D GPS lock; Numata=10 means 10 satellites are visible (a healthy number for SITL); voltage=12.587 is a fully charged simulated 3S battery; level=100 means 100% charge; and 0.92 seconds since the last heartbeat confirms a live, healthy connection.